

Introduction

Programming for Economists

Jeppe Druedahl

1. Intended learning goals
2. Programming in practice
3. Infrastructure
4. Work-flow
5. Projects
6. Summing up

Intended learning goals

- **In a nutshell:**

1. Summarize and visualize empirical data
 - ⇒ gain flexibility and automation compared to Excel
 - ⇒ build upon Descriptive Economics
 - ⇒ working with »big data«
2. Numerically solve and simulate economic models
 - ⇒ less strict assumptions than with math and easy visualization
 - ⇒ improve your understanding of Micro and Macro
 - ⇒ from model-user to *model-builder*

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- **Tool:** *Python* - general purpose programming language (open source)

1. Web development (not covered)
2. Data science and machine learning (covered)
3. Scientific computing and research (main focus)

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 - We give you a broad overview
 - We give you set of important tools
 - You have to experiment on your own
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- **High level:** Few (if any) econ bachelor programs provide education on programming and numerical analysis on the level you will get

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Scientific programming

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- **Programming is more than writing code:** Structuring, testing, documenting and collaborating on code projects is a central aspect of this course

You and your teachers

- **Jeppe Druedahl**, Associate Professor
web: sites.google.com/view/jeppe-druehdahl/
e-mail: jeppe.druehdahl@econ.ku.dk
- **Ongoing research:** Using »deep learning« to solve a large life-cycle model estimated on Danish individual-level data



Second time

- **Previous course:**

 - Introduction to Programming and Numerical Analysis

1. Higher level
2. Almost fully video-based (few physical lectures)

- **Second time for this course**

1. *All of your feedback is very important for improving the course!*
(since 1st time: restructured and simpler economic problems)
2. Now an obligatory course on the 3rd semester
3. Goal is to be more video-based in future

Programming in practice

Python distribution and VSCode

- We work with **Python 3.14**
- **Suggested environment:**
 1. **Distribution:** Anaconda
 2. **Editor/IDE:** VSCode (with free copilot)
- **Example:** Show how to use *VSCode*
 1. Run *python code* and *notebooks*
 2. Import US GDP data from online API and plot
 3. Solve the consumer problem from microeconomics

Infrastructure

Getting started

- **Web-page:** The course is organized around <https://sites.google.com/view/numeconcp-ph-progecon/>
- **DataCamp:** Online courses on Python (requires no installation)
⇒ you get 6 months free access (see e-mail with details)
- **Installation of Python:** Follow the [installation guide](#)

GitHub.com (code hosting platform)

- All course materials will be shared on GitHub
- Organization: www.github.com/NumEconCopenhagen

Repositories:

1. **ProgEcon-lectures:** slides, course plan, guides etc.
 2. **ProgEcon-exercises:** problem sets, solutions etc.
- Need-to-now: Click on »<> Code« and »Download zip«
 - **Git** (more on this later): A version-control system for tracking changes in files and coordinating work ⇒ integrated in VSCode

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- **Exam requirements** (see **deadlines**)
 1. Basic programming test (on **DataCamp.com**, see e-mail)
 2. Project 1: Data project
 3. Project 2: Model project
- **Exam:** Portfolio of revised projects (40%)
+ 36 hours home assignment (60%)
- **Grading:** Pass or fail
- **Groups:** All projects can be done in *fixed* groups (maximum of 4)

Course plan - lectures

1. **Basics** (types, containers, views vs. copies, conditionals, loops, functions, floating point numbers, classes, numpy, print, plot)
2. **Tools** (optimization, root-finding, random numbers, simulation)
3. **Data** (save/load, online data, descriptive economics)
4. **Models** (Solow, Walras, AS-AD, calibration/estimation, structure and documentation, workflow and debugging, git)
5. **Perspectives** (algorithms, speed, parallization, dynamic programming, artificial intelligence)

Lectures are recorded (and uploaded raw without editing)

More screen capture videos from *Introduction to Programming and Numerical Analysis* are extra material

Course plan - exercises

1. Problem Set 0: Getting started + DataCamp
2. Problem Set 0: Getting started + DataCamp
3. Problem Set 0: Getting started + DataCamp
4. Problem set 1. Printing and Plotting
5. Problem set 2. Optimization
6. Problem set 3: Random Numbers
7. Problem set 4. Descriptive Economics I
8. Problem set 5. Descriptive Economics II + Data Project
9. Work on Data Project
10. Problem set 6: Solow
(hand-in data project)
11. Problem set 7: Walras
12. Problem set 8: AS-AD (+ Data Project feedback)
13. Model Project (+ intro to git)
14. Work on Model Project (+ exam preparation)
(hand-in model project)

Receive written feedback om Model Project) + Exam

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3. **Models:** Hard to understand math and code simultaneously.
But the combination is really powerful.
4. **Perspectives:** I have learned a lot! And so many possibilities ahead for working with data and models.

Work-flow

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- **In between classes and lectures:**

1. Go through lecture notebooks (curriculum)
2. Solve the problem set
3. Experiment with your own ideas

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- **Help each other!!** You will learn a lot.
Remember to be constructive and polite!

Projects

Basic programming test

- **You must complete the following courses on DataCamp**

Required:

1. Introduction to Python
2. Intermediate Python

Optional:

1. Introduction to Functions in Python
2. Introduction to NumPy
3. Intermediate Object-Oriented Programming in Python
(chapter: Overloading and Multiple Inheritance)

- **First 3 exercise classes:** Reserved for your work on DataCamp

Data project

- **Objectives:**

1. Apply data cleaning and data structuring methods
2. Apply data analysis methods
3. Structure a code project
4. Document code
5. Present results in text form and in figures

- **Content:**

1. Import data from an online source
2. Present the data visually
3. Apply some method(s) from descriptive economics

- **Structure:**

1. A self-contained single notebook presenting the analysis
2. Fully documented python files

- **Hand-in:** Upload on Absalon

Model project

- **Objectives:**

1. Apply model analysis methods
2. Structure a code project
3. Document code
4. Present results in text form and in figures

- **Content:**

1. Describe an algorithm on how to solve a simple economic model
2. Solve (and perhaps simulate) a simple economic model
3. Visualize results across e.g. parametrizations
4. Analyze one or more extensions of the baseline model

- **Structure:**

1. A self-contained single notebook presenting the analysis
2. Fully documented python files

- **Hand-in:** Upload on Absalon

Summing up

- **I hope you have an idea of:**

1. Why learning programming and numerical analysis is important
2. What you will learn in this course
3. Why active participation is required
4. How you will qualify for and pass the exam

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3. **Third priority:** Download lecture and exercise material
4. **Fourth priority:** Run the example code from this lecture yourself